

MENDOZA, ANDREA GAILLE D.

BSCS - C302

MySQL Workbench - CALL CENTER - Warning - n...

File Edit View Query Database Server Tools Scripting Help

Navigation: **SCHEMAS** Filter objects: **project** Tables Views Stored Procedures Functions test

Administration: **Schemas** Connection Details: Name: CALL CENTER Host: 127.0.0.1 Port: 3306 Login User: root Current: root@localhost SSL cipher: SSL not used Server: mariadb.org Product: binary distribution Version: 10.4.32-MariaDB Connector: C++ 9.6.0

SQL Editor: **calls**

```
21 SET call_timestamp = STR_TO_DATE(call_timestamp, '%m/%d/%Y');
22
23 SET SQL_SAFE_UPDATES = 1;
24
25 SET SQL_SAFE_UPDATES = 0;
26
27 UPDATE calls
28 SET csat_score = NULL
29 WHERE csat_score = '0';
30
31 SET SQL_SAFE_UPDATES = 1;
32
33 SELECT * FROM calls LIMIT 10;
34
35 SELECT COUNT(*) AS rows_num FROM calls;
36 SELECT COUNT(*) AS cols_num FROM information_schema.columns WHERE table_name = 'calls';
37
38 SELECT DISTINCT sentiment FROM calls;
```

Result Grid: 1000 rows

ID	cust_name	sentiment	csat_score	call_timestamp	reason	city	state	channel	response_time	call_duration_minutes	call_center	
14	006-5707809-0554614J	customer_name	Neutral	7	10/29/2020	Billing Question	Detroit	Michigan	Call-Center	Within SLA	17	Los Angeles,CA
	Q06-72219678-10213947	Cochran Kobbly	Very Positive	0	10/05/2020	Service Outage	Spartanburg	South Carolina	ChatBot	Within SLA	23	Baltimore,MD
	012-5022802-A-0281914D	Award Brandshet	Negative	0	10/04/2020	Billing Question	Gainesville	Florida	Call-Center	Above SLA	45	Los Angeles,CA
	218-96487559-430008-e7	Norren LaRfina	Very Negative	1	10/17/2020	Billing Question	Portland	Oregon	ChatBot	Within SLA	12	Los Angeles,CA
	026-58462759-0-12846276	Tonia Van der Bellen	Very Positive	0	10/17/2020	Payment	Fort Wayne	Indiana	Call-Center	Within SLA	23	Los Angeles,CA
	301-79728660-0-224355-4e	Kaylen Eriien	Neutral	5	10/20/2020	Billing Question	Salt Lake City	Utah	Call-Center	Within SLA	25	Baltimore,MD
	A2Z-6934097-e-3834247T	Philoe Bowring	Neutral	8	10/16/2020	Billing Question	Tyler	Texas	ChatBot	Within SLA	31	Baltimore,MD
	T00-2707918-1-4087893-3e	Krysia de Tocuquerfle	Positive	0	10/21/2020	Billing Question	New York City	New York	ChatBot	Below SLA	37	Los Angeles,CA
	XNG-44599118-P-5444753J	Oran Lifaey	Very Negative	0	10/05/2020	Billing Question	Dallas	Texas	Email	Below SLA	37	Baltimore,MD

calls 19 x

Output: Action Output

#	Time	Action	Message	Duration / Freq
1	10:41:41	SELECT COUNT(*) AS rows_num FROM calls LIMIT 10	1 rows returned	0.016 sec / 0
2	10:41:41	SELECT COUNT(*) AS cols_num FROM information_schema.columns WHERE table_name = 'calls' LIMIT 0, 1000	1 rows returned	0.000 sec / 0

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

NUMBER OF COLUMNS

MySQL Workbench - CALL CENTER - Warning - n...

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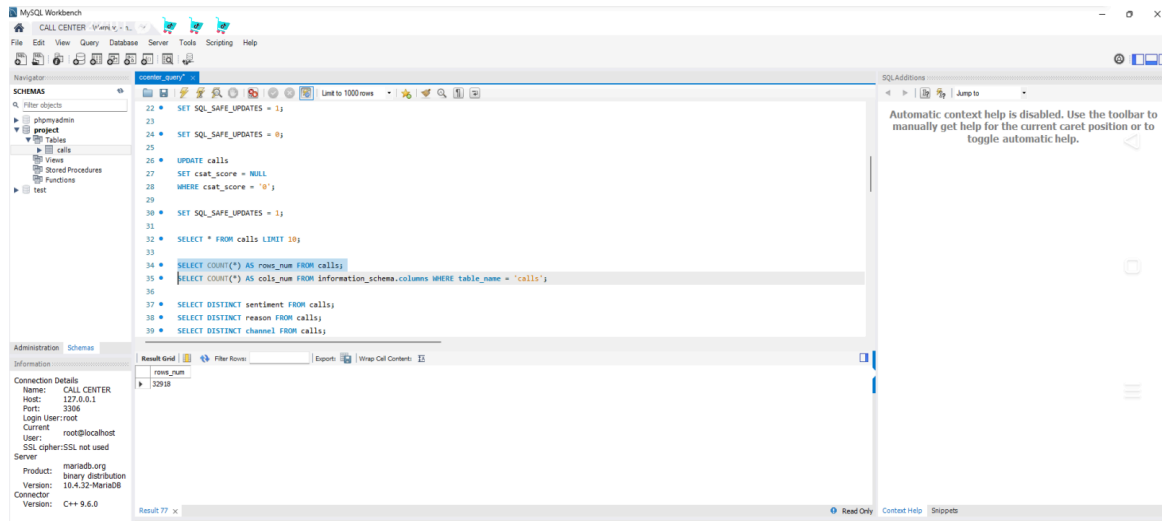
calls 19 x

Output: Action Output

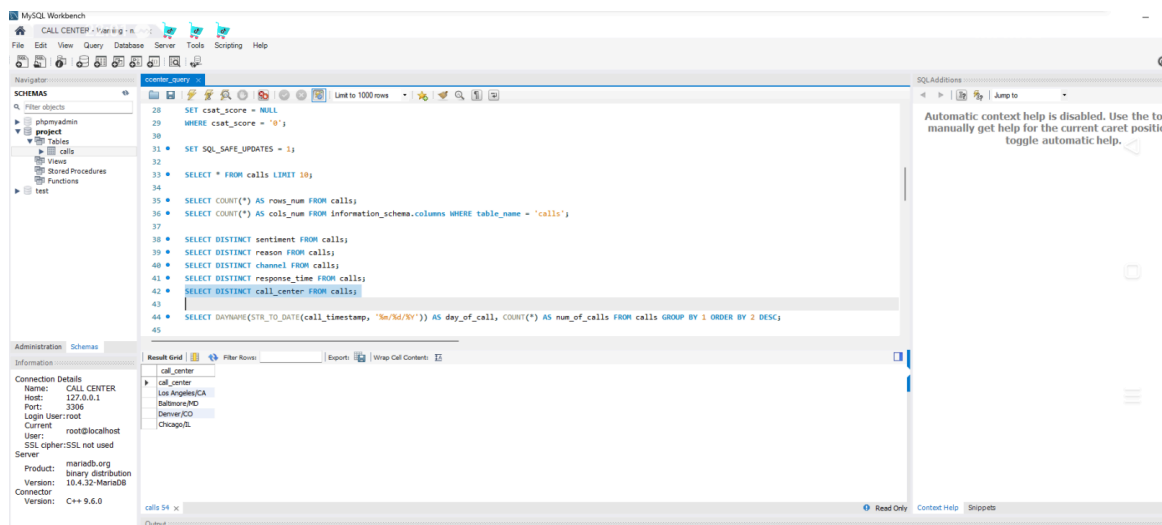
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NUMBER OF ROWS



DIFFERENT VALUES POSSIBLE FOR THE ROWS WE SELECTED



DAY HAS THE MOST CALLS

The screenshot shows the MySQL Workbench interface. The left sidebar displays the 'SCHEMAS' panel with a tree view showing 'project' and 'test' databases. The main editor window contains a SQL query that uses a subquery to find the day with the most calls and then calculates the percentage of total calls for each day. The query is as follows:

```
39 * SELECT DISTINCT channel FROM calls;
40 * SELECT DISTINCT response_time FROM calls;
41 * SELECT DISTINCT call_center FROM calls;
42
43 * SELECT DAYNAME(STR_TO_DATE(call_timestamp, '%m/%d/%Y')) AS day_of_call, COUNT(*) AS num_of_calls FROM calls GROUP BY 1 ORDER BY 2 DESC;
44
45 * SELECT sentiment, count(*), ROUND((COUNT(*) / (SELECT COUNT(*) FROM calls)) * 100, 1) AS pct
46 FROM calls GROUP BY 1 ORDER BY 3 DESC;
47
48 * SELECT reason, count(*), ROUND((COUNT(*) / (SELECT COUNT(*) FROM calls)) * 100, 1) AS pct
49 FROM calls GROUP BY 1 ORDER BY 3 DESC;
50
51 * SELECT channel, count(*), ROUND((COUNT(*) / (SELECT COUNT(*) FROM calls)) * 100, 1) AS pct
52 FROM calls GROUP BY 1 ORDER BY 3 DESC;
53
54 * SELECT response_time, count(*), ROUND((COUNT(*) / (SELECT COUNT(*) FROM calls)) * 100, 1) AS pct
55 FROM calls GROUP BY 1 ORDER BY 3 DESC;
```

The 'Result Grid' shows the results of the query, with columns 'day_of_call' and 'num_of_calls'. The results are as follows:

day_of_call	num_of_calls
Thursday	5543
Friday	5512
Monday	4535
Saturday	4464
Wednesday	4480
Tuesday	4284
Sunday	4124
2008	6

The 'Output' panel shows the execution progress of the query, with messages indicating the number of rows returned for each step.

QUERYING THE MIN, MAX AND THE AVERAGE CALL DURATION IN MINUTES

The screenshot shows the MySQL Workbench interface. The left sidebar displays the 'SCHEMAS' panel with a tree view showing 'project' and 'test' databases. The main editor window contains a SQL query that uses a subquery to find the min, max, and average call duration in minutes. The query is as follows:

```
63 FROM calls GROUP BY 1 ORDER BY 2 DESC;
64
65 * SELECT MIN(csat_score), MAX(csat_score), ROUND(AVG(csat_score),1)
66 FROM calls WHERE csat_score != 0;
67
68 * SELECT MIN(call_timestamp), MAX(call_timestamp)
69 FROM calls;
70
71 * SELECT MIN(call_duration_minutes), MAX(call_duration_minutes), AVG(call_duration_minutes)
72 FROM calls;
73
74 * SELECT call_center, response_time, COUNT(*)
75 FROM calls GROUP BY 1,2 ORDER BY 3,3 DESC;
76
77 * SELECT call_center, AVG(call_duration_minutes)
78 FROM calls GROUP BY 1 ORDER BY 2 DESC;
79
```

The 'Result Grid' shows the results of the query, with columns 'MIN(call_duration_minutes)', 'MAX(call_duration_minutes)', and 'AVG(call_duration_minutes)'. The results are as follows:

MIN(call_duration_minutes)	MAX(call_duration_minutes)	AVG(call_duration_minutes)
0	45	25.0885

The 'Output' panel shows the execution progress of the query, with messages indicating the number of rows returned for each step.

SHOWS HOW MANY CALLS ARE WITHIN, BELOW OR ABOVE THE SLA TIME

The screenshot shows a MySQL Workbench interface with a query window titled 'center_query'. The query is as follows:

```
63 FROM calls GROUP BY 1 ORDER BY 2 DESC;
64
65 * SELECT MIN(csat_score), MAX(csat_score), ROUND(AVG(csat_score),1)
66 FROM calls WHERE csat_score != 0;
67
68 * SELECT MIN(call_timestamp), MAX(call_timestamp)
69 FROM calls;
70
71 * SELECT MIN(call_duration_minutes), MAX(call_duration_minutes), AVG(call_duration_minutes)
72 FROM calls;
73
74 * SELECT call_center, response_time, COUNT(*)
75 FROM calls GROUP BY 1,2 ORDER BY 1,3 DESC;
76
77 * SELECT call_center, AVG(call_duration_minutes)
78 FROM calls GROUP BY 1 ORDER BY 2 DESC;
79
```

The Results Grid shows the following data:

call_center	response_time	COUNT(*)
Baltimore.MD	Within SLA	6725
Baltimore.MD	Below SLA	2730
Baltimore.MD	Above SLA	1118
call_center	response_time	6
Chicago.IL	Within SLA	3652
Chicago.IL	Below SLA	1262
Chicago.IL	Above SLA	642
Denver.CO	Within SLA	1767
Denver.CO	Below SLA	759
Denver.CO	Above SLA	286
Los Angeles...	Within SLA	8580

The Administration panel on the left shows connection details for 'CALL CENTER'.

WINDOW FUNCTION

The screenshot shows a MySQL Workbench interface with a query window titled 'center_query'. The query is as follows:

```
84 FROM calls GROUP BY 1 ORDER BY 2 DESC;
85
86 * SELECT state, reason, COUNT(*)
87 FROM calls GROUP BY 1,2 ORDER BY 1,3 DESC;
88
89 * SELECT state, sentiment, COUNT(*)
90 FROM calls GROUP BY 1,2 ORDER BY 1,3 DESC;
91
92 * SELECT state, AVG(csat_score)
93 FROM calls WHERE csat_score != 0 GROUP BY 1 ORDER BY 2 DESC;
94
95 * SELECT sentiment, AVG(call_duration_minutes)
96 FROM calls GROUP BY 1 ORDER BY 2 DESC;
97
98 * SELECT call_timestamp, MAX(call_duration_minutes) OVER(PARTITION BY call_timestamp) AS max_call_duration
99 FROM calls ORDER BY 2 DESC;
100
```

The Results Grid shows the following data:

call_timestamp	max_call_duration
10/10/2020	45
10/26/2020	45
10/12/2020	45
10/03/2020	45
10/20/2020	45
10/26/2020	45
10/24/2020	45
10/13/2020	45
10/01/2020	45
10/26/2020	45
10/25/2020	45

The Output panel shows the following messages:

```
148 22:28:51 SELECT DAYNAME(STR_TO_DATE(call_timestamp, '%m/%d/%Y')) AS day_of_cal, COUNT(*) AS num_of_calls FROM calls GROUP BY 1 ORDER BY 1; 8 row(s) returned
149 22:28:51 SELECT DAYNAME(STR_TO_DATE(call_timestamp, '%m/%d/%Y')) AS day_of_cal, COUNT(*) AS num_of_calls FROM calls GROUP BY 1 ORDER BY 1; 8 row(s) returned
```